Chapter 1

Computer Abstractions and Technology

How Can We Further Improve Performance?

CPU Time = Instruction Count × CPI × Clock Cycle Time

What can processor designers change to keep improving performance?

Exploit

Use Multiple

Q1. Which of the following represents a different strategy from simply other choices?

A. **Increasing the clock frequency**

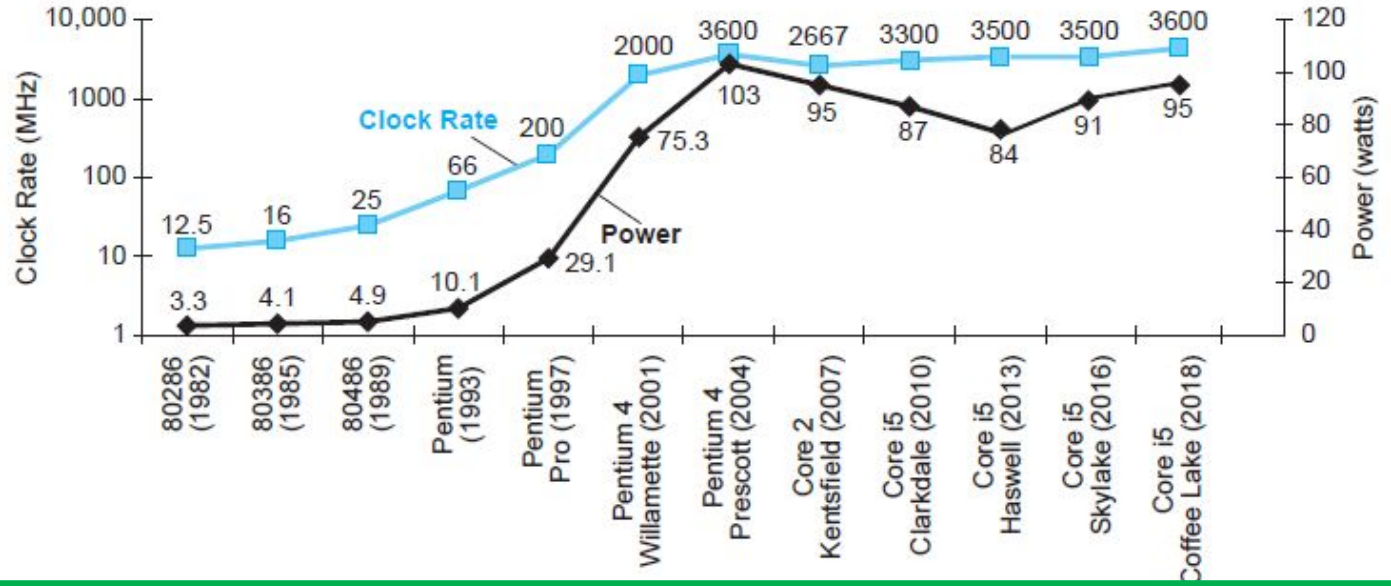
B. **Executing multiple instructions within the same time period**

C. **Shortening the clock cycle time**

D. **Increasing the supply voltage**

A1. B

Power Trends



Q2. Clock frequency increased dramatically over time. Why didn't power consumption increase at the same rate?

A2. Because supply voltage was significantly reduced, which helped offset the increase in power caused by higher frequency and capacitive load.

Reducing Power

- Suppose a new CPU has
  - 85% of capacitive load of old CPU
  - 15% voltage and 15% frequency reduction

$$\frac{P_{new}}{P_{old}} = \frac{C_{old} \times 0.85 \times (V_{old} \times 0.85)^2 \times F_{old} \times 0.85}{C_{old} \times V_{old}^2 \times F_{old}} = 0.85^4 = 0.52$$

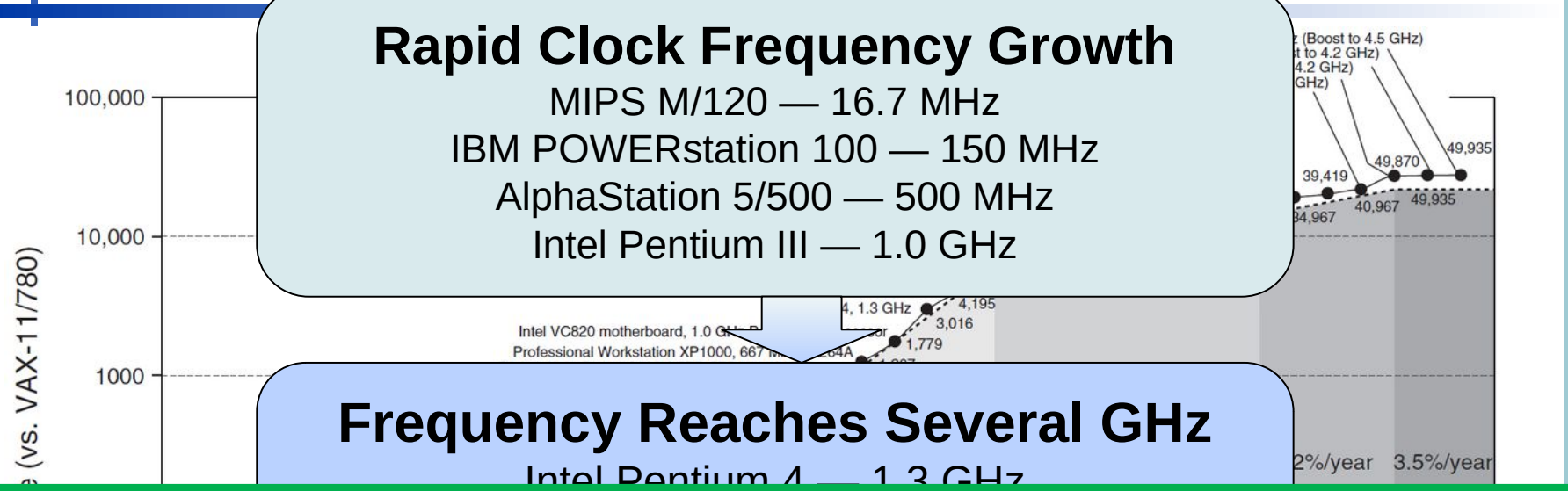
Q3. A processor increases its clock frequency by 25%. Assume the capacitive load remains unchanged. By approximately what percentage must the voltage be reduced to keep the same power consumption?

A3. P_old / P_new = 1.25 * (V_old / V_new)^2=1
V_new / V_old ≈ 0.894 => reduction ≈ 10.6%

Uniprocessor Performance

Rapid Clock Frequency Growth

MIPS M/120 — 16.7 MHz
IBM POWERstation 100 — 150 MHz
AlphaStation 5/500 — 500 MHz
Intel Pentium III — 1.0 GHz



Frequency Reaches Several GHz

Intel Pentium 4 — 1.3 GHz

Q4. Which trend best shows a shift in processor performance strategy?

A. **Clock frequency increased from MHz to GHz.**

B. **Later processors used more cores at similar GHz frequencies.**

C. **All processors reached the same clock frequency.**

D. **Multicore processors always used lower frequencies.**

A4. B

Multiprocessors

- Multicore microprocessors
  - More than one processor per chip
- Requires explicitly parallel programming
  - Compare with instruction level parallelism

Q5. Which statement best distinguishes multicore from ILP?

A. Multicore uses multiple cores; ILP exploits parallelism within a core.

B. Multicore increases clock rate; ILP reduces clock rate.

C. Multicore is hidden from programmers; ILP requires parallel programming.

D. They are exactly the same technique.

A5. A

SPEC CPU Benchmark

- Programs used to measure performance

Q6. A processor has execution time ratios of 2.0, 1.0, and 4.0 for three benchmarks. What is the geometric mean?

A6. 2.0

  - Elapsed time to execute a selection of programs
    - Negligible I/O, so focuses on CPU performance
  - Normalize relative to reference machine
  - Summarize as geometric mean of performance ratios
    - CINT2006 (integer) and CFP2006 (floating-point)

$$\sqrt[n]{\prod_{i=1}^{n} \text{Execution time ratio}_i}$$

Pitfall: MIPS as a Performance Metric

- MIPS: Millions of Instructions Per Second
  - Doesn't account for

Q7. Two ISAs execute the same task. Which one is faster?
ISA A: 1M instructions, 2 µs per instruction
ISA B: 4M instructions, 1 µs per instruction
Which ISA completes the task faster?

A7. A is faster, even if each instruction takes longer.
0.5 MIPS vs. 1 MIPS

ISA A — Complex instruction	ISA B — Simpler instructions
ADD C, A, B	LOAD R1, A LOAD R2, B ADD R3, R1, R2 STORE C, R3

Pitfall: Amdahl's Law

- Improving an aspect of a computer and expecting a proportional improvement in overall performance

$$T_{improved} = \frac{T_{affected}}{improvement\ factor} + T_{unaffected}$$

Q8. A program spends 75% of its execution time on a part that can be accelerated by 5×. The remaining 25% is unchanged. What is the overall speedup?

A8.

The accelerated part now takes:

75/5=15%

The new execution time is:15% + 25% = 40%

Therefore: Overall Speedup = 100/40 = 2.5

Concluding Remarks

- Cost/performance is improving
  - Due to underlying technology development
- Hierarchical layers of abstraction
  - In both hardware and software
- Instruction set architecture
  - The hardware/software interface
- Execution time: the best performance measure
- Power is a limiting factor
  - Use parallelism to improve performance

Section 1-4 Quick Questions

Q1. Which strategy differs from the other two: higher clock rate, ILP, or multicore?

Q2. Why did voltage scaling help control CPU power?

Q3. If clock rate rises by 25%, how much must voltage decrease to keep power constant?

Q4. What processor trend shows the shift toward multicore?

Section 1-4 Quick Questions

Q5. What is the key difference between ILP and multicore?

Q6. Find the geometric mean of the ratios 2.0, 1.0, and 4.0.

Q7. ISA A: 1M instructions × 2 μs
ISA B: 4M instructions × 1 μs
Which ISA is faster?

Q8. 75% of a program is accelerated by 5×.
What is the overall speedup?

Exercise 1 | Power Consumption

A processor is redesigned with 80% of the original capacitive lo ad, 90% of the original voltage, and 120% of the original clock fr equency.

Calculate P_new / P_old. Is the new dynamic power higher or lower?

Answer

$P_{new} / P_{old} = 0.8 \times (0.9)^2 \times 1.2 = 0.7776$
$P_{new} \approx 77.8\%$ of P_{old}

Dynamic power decreases by ≈ 22.2%.

Key Point: Power is proportional to C and f, but to V^2 ; voltage scali ng has a particularly strong effect on power.

Exercise 2 | Voltage and Clock Frequency

A processor increases its clock frequency by 44% while its ca pacitive load remains unchanged.

By what percentage must the voltage be reduced to keep the same dynamic power?

Answer

$1.44 \times (V_{new} / V_{old})^2 = 1$
$V_{new} / V_{old} = \sqrt{(1 / 1.44)} = 0.8333$
Voltage reduction $= 1 - 0.8333 \approx 16.7\%$

Key Point: When clock rate increases, voltage must decrease to maintain the same power, illustrating a key limitation behind the Power Wall.

Exercise 3 | SPEC Benchmark

A reference machine takes 40 s, 90 s, and 160 s for three benchm arks.
A new processor takes 20 s, 30 s, and 40 s.

Calculate the three performance ratios and their geometric mean.

Answer

$R_1 = 40 / 20 = 2$ $R_2 = 90 / 30 = 3$ $R_3 = 160 / 40 = 4$
Geometric Mean = $\sqrt[3]{(2 \times 3 \times 4)} = \sqrt[3]{24} \approx 2.88$

Key Point: SPEC normalizes each benchmark to a reference machine and uses the geometric mean to summarize overall performance.

Exercise 4 | MIPS Pitfall

Processor A executes 600M instructions in 0.4 s.
Processor B executes 1000M instructions in 0.5 s for the same task.

Calculate the MIPS of each processor. Which processor has better performance?

Answer

MIPS_A = 600 / 0.4 = 1500 MIPS
MIPS_B = 1000 / 0.5 = 2000 MIPS
But: T_A = 0.4 s < T_B = 0.5 s
Processor A has better performance.

Key Point: Higher MIPS does not necessarily mean higher performance; execution time should be used for the same workload.

Exercise 5 | Amdahl's Law and Multicore

80% of a program can be parallelized. On four cores, assume t he parallel portion achieves an ideal 4× speedup; the remainin g 20% is unchanged.

(a) Calculate the overall speedup.

(b) What is the maximum speedup as the number of cores approaches infinity?

Answer

(a) New time = 0.20 + 0.80 / 4 = 0.40

Overall speedup = 1 / 0.40 = 2.5×

Maximum speedup = 1 / 0.20 = 5×

(b) As cores → ∞: New time → 0.20

Key Point: Multicore speeds up only the parallelizable portion; the serial portion limits the maximum overall speedup.